

Lecture-5: Applications Transformer and DC Machine

Energy Conversion

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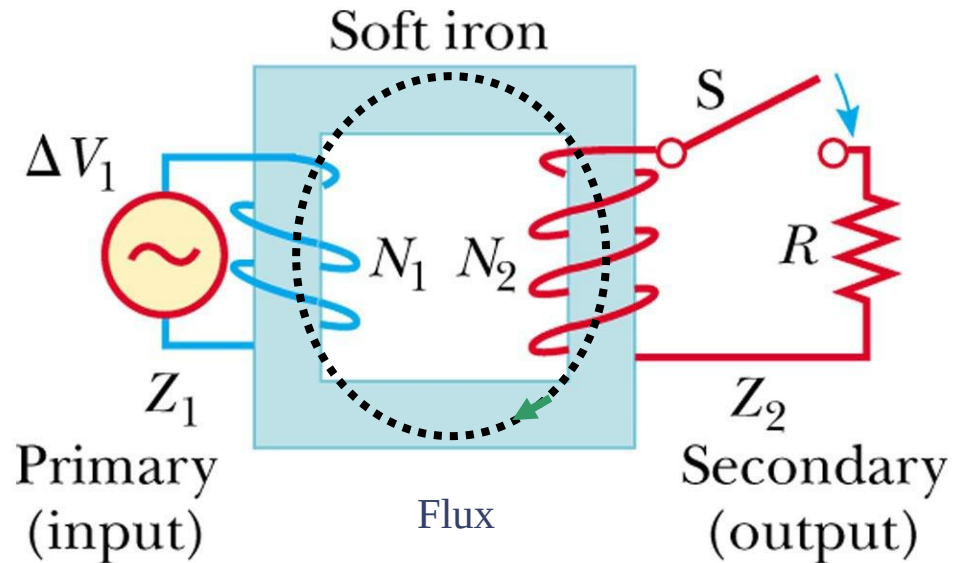
Outlines

- Structure
- Physics
- Ideal transformer
- Practical transformer
- Equivalent circuits

- Objectives
 - Understanding of physics of energy conversion
 - Simple modeling and analysis

A Conceptual Transformer

- An simple AC transformer consists of two coils of wire wound around a core of soft iron
- The side connected to the input AC voltage source is called the *primary* and has N_1 turns
- The side connected to the “load” side is called the *secondary* and has N_2 turns.
- Both currents magnetize the core only the primary does energization/excitation, meaning providing input energy



$$\text{Turn ratio } (a) = N_1 / N_2$$

How Does It Work

Current from the source generates a magnetic field,

Magnetic field magnetizes the media to produce flux

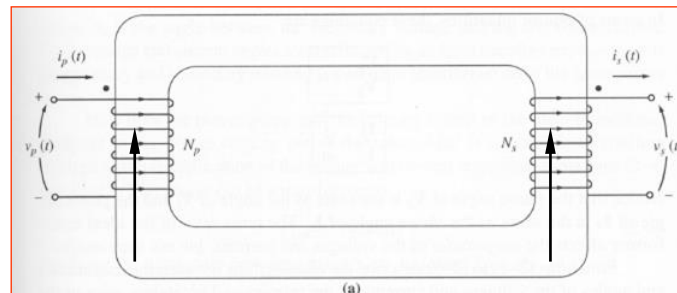
Changing current will induce a changing flux

A changing flux induces voltage potential, and current in the secondary windings if it is closed. => so alternating sources is needed.

Roles of the Primary and the Secondary

A two-winding transformer consists of two windings interlinked by a mutual magnetic field.

- Primary winding
 - The core is energized by connecting it to an input source: excitation
 - Sending out power
- Secondary winding – winding to which an electrical load is connected
 - from which output energy is drawn.



Primary winding

Secondary winding

Voltage Relation

- Now the primary winding and second winding share the same interlinked flux

$$\begin{aligned} v_S(t) &= N_S \frac{d\phi}{dt} \\ \Rightarrow v_P(t) &= N_P \frac{d\phi}{dt} \\ \frac{v_S(t)}{v_P(t)} &= \frac{N_S}{N_P} \quad \text{and of course} \quad \frac{V_{rms-S}}{V_{rms-P}} = \frac{N_S}{N_P} \end{aligned}$$

Current Ratio

- Since the reluctance of a well-designed transformer core is very small (close to zero) until the core is saturated, the relationship between the primary and secondary currents is approximately...

$$\mathfrak{I}_{net} = N_P i_P - N_S i_S = \phi \mathfrak{R} \approx 0 \quad \because \mathfrak{R} = 0$$

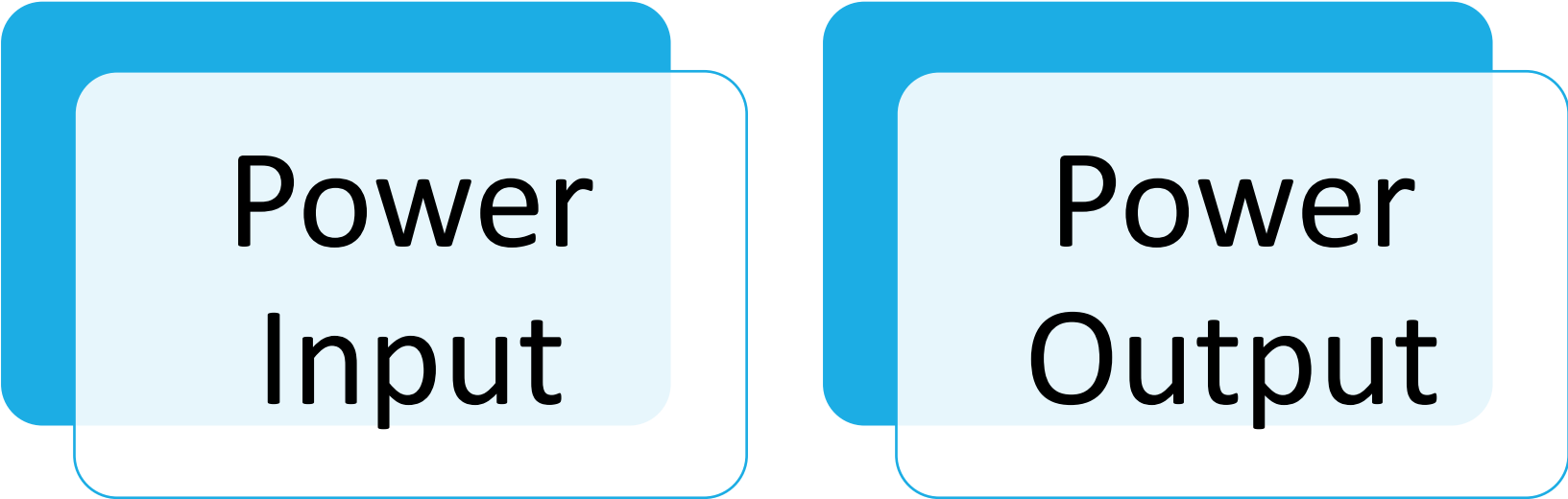
$\mathfrak{R}$ is very small for ferromagnetic material with high permeability

- According to Lenz's Law, current must flow into on dotted end and out of the other dotted end.

$$N_P i_P \approx N_S i_S \quad \text{or} \quad \frac{i_P}{i_S} \approx \frac{N_S}{N_P} = \frac{1}{a}$$



Power Relation



The diagram consists of two identical rectangular boxes side-by-side. Each box has a thick blue top bar and a light blue rounded rectangle in the center. The text 'Power Input' is centered in the left box, and 'Power Output' is centered in the right box.

**Power
Input**

**Power
Output**

Factors to Be Considered in Practical Transformer

- Saturation and Hysteresis effects are ignored in normal operating condition
- Leakages
- Eddy and excitation currents

Impact of Transformer Structure: Leakages of Flux

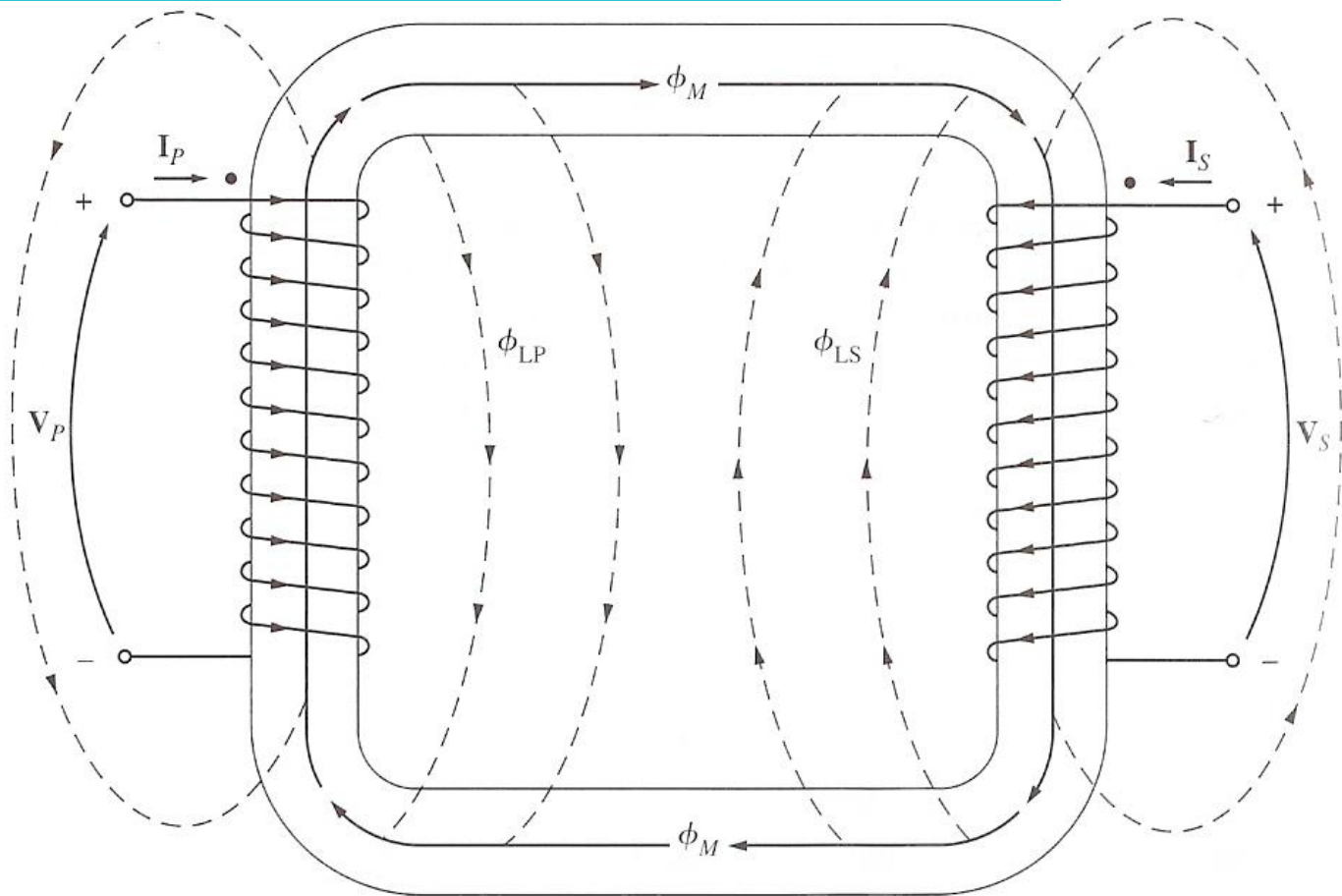
- Average flux per turn can be defined as $\bar{\phi} = \frac{\lambda}{N}$ and Faraday's law can be written as $e_{ind} = \frac{d\lambda}{dt} = N \frac{d\bar{\phi}}{dt}$.
- $\bar{\phi} = \frac{1}{N_p} \int v_p(t) dt$ states that the average flux in the winding is proportional to the integral of the voltage applied to the winding, and the constant of proportionality is $\frac{1}{N_p}$.

- When leakage of winding is considered,
 - The flux in the primary coil of the transformer has two components: Mutual flux and Leakage flux

$$\bar{\phi}_P = \phi_M + \phi_{LP}$$

- In the secondary coil, the flux also has mutual flux and leakage flux.

$$\bar{\phi}_S = \phi_M + \phi_{LS}$$



Impact of Leakages on Voltages

- The voltage on the primary coil of transformer can be expressed in terms of Faraday's law as

$$v_P(t) = N_P \frac{d\phi_P}{dt} = N_P \frac{d\phi_M}{dt} + N_P \frac{d\phi_{LP}}{dt} = e_P(t) + e_{LP}(t)$$

- In the secondary coil, the voltage also can be expressed as

$$v_S(t) = N_S \frac{d\phi_S}{dt} = N_S \frac{d\phi_M}{dt} + N_S \frac{d\phi_{LS}}{dt} = e_S(t) + e_{LS}(t)$$

- The primary voltage due to the mutual flux $e_P(t) = N_P \frac{d\phi_M}{dt}$ and the secondary voltage due to the mutual flux $e_S(t) = N_S \frac{d\phi_M}{dt}$

result in the relationship of $\frac{e_P(t)}{N_P} = \frac{d\phi_M}{dt} = \frac{e_S(t)}{N_S}$

- Therefore, $\frac{e_P(t)}{e_S(t)} = \frac{N_P}{N_S} = a$, it means that the ratio of the primary voltage caused by the mutual flux to the secondary voltage caused by the mutual flux is equal to the turns ratio of the transformer.

Ideal Transformer

- The smaller the leakage fluxes of the transformer are, the closer the total transformer voltage ratio approximates that of the ideal transformer.

$$\phi_M \gg \phi_{LP} \text{ and } \phi_M \gg \phi_{LS} \quad \rightarrow \quad \frac{v_P(t)}{v_S(t)} = \frac{N_P}{N_S} = a$$

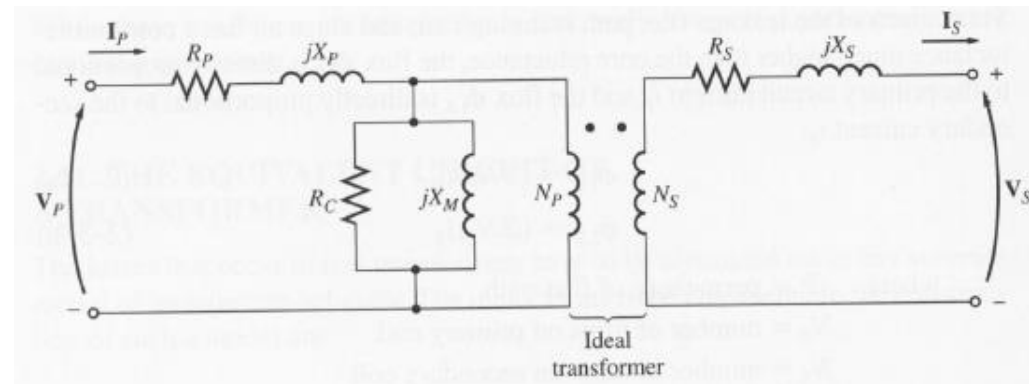
Equivalent Circuit of Practical Transformer

Modeling the copper losses: resistive losses in the primary and secondary windings of the core, represented in the equivalent circuit by R_p and R_s .

Modeling the leakage fluxes: primary leakage flux is proportional to the primary current I_p and secondary leakage flux is proportional to the secondary current I_s , ϕ_{LP} represented in the equivalent circuit by $L_p (= N_p \phi_{LP} / I_p)$ and $L_s (= N_s \phi_{LP} / I_s)$ (because of the definition $L = N \phi / I$)

Modeling the core excitation: I_m is proportional to the voltage applied to the core and lags the applied voltage by 90° . It is modeled by X_M .

Modeling the core loss current: I_{h+e} is proportional to the voltage applied to the core and in phase with the applied voltage. It is modeled by R_C .



The Equivalent Circuit

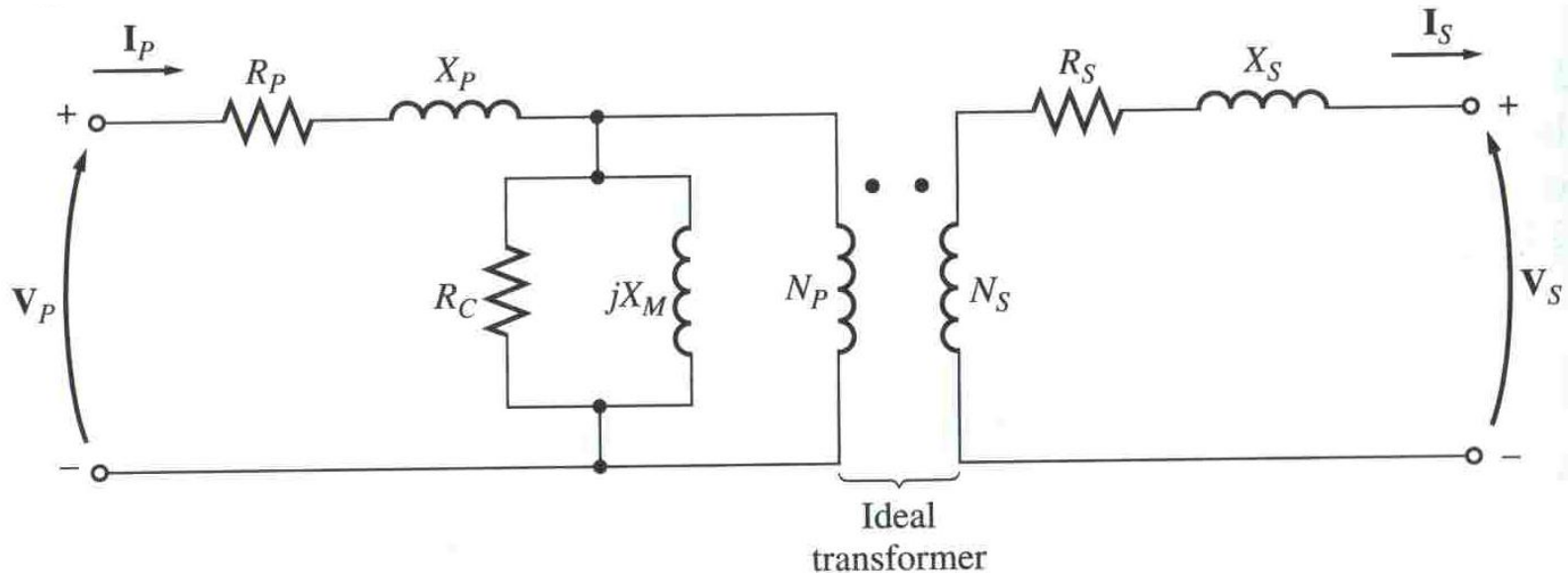
- Copper(I^2R) losses are resistive losses in the primary and secondary windings of the transformer core. They are modeled by placing a resistor R_p in the primary circuit of the transformer and a resistor R_s in the secondary circuit.
- Leakage fluxes are modeled by primary and secondary inductors.
- The magnetization current i_m is a current proportional (in the unsaturated region) to the voltage applied to the core and lagging the applied voltage by 90° . So it can be modeled by a reactance X_M connected across the primary voltage source.
- The core-loss current i_{h+e} is a current proportional to the voltage applied to the core that is in phase with the applied voltage, so it can be modeled by a resistance R_c connected across the primary voltage source.

The Equivalent Circuit of a Real Transformer

The losses that occur in transformers have to be accounted for in any accurate model of transformer behavior.

1. *Hysteresis losses and Saturation.*
2. *Leakage flux.* The fluxes which escape the core and pass through only one of the transformer windings are leakage fluxes. These escaped fluxes produce a self-inductance in the primary and secondary coils, and the effects of this inductance must be accounted for.
3. *Copper (I^2R) losses.* Copper losses are the resistive heating losses in the primary and secondary windings of the transformer. They are proportional to the square of the current in the windings.
4. *Eddy current losses.* Eddy current losses are resistive heating losses in the core of the transformer. They are proportional to the square of the voltage applied to the transformer.

The Equivalent Circuit



Variables at the primary side

Variables at the secondary side

Power in Ideal Transformer

Real power P supplied to the transformer by the primary circuit

$$P_{in} = V_p I_p \cos \theta_p$$

$$\theta_p = \theta_s = \theta$$

Real power coming out of the secondary circuit

$$P_{out} = V_s I_s \cos \theta_s = \left(\frac{V_p}{a} \right) (a I_p) \cos \theta = V_p I_p \cos \theta = P_{in}$$

Thus, *the output power of an ideal transformer is equal to its input power.*

The same relationship applies to reactive Q and apparent power S :

$$Q_{in} = V_p I_p \sin \theta = (a V_s) \left(\frac{I_s}{a} \right) \sin \theta = V_s I_s \sin \theta = Q_{out}$$

$$S_{in} = V_p I_p = V_s I_s = S_{out}$$

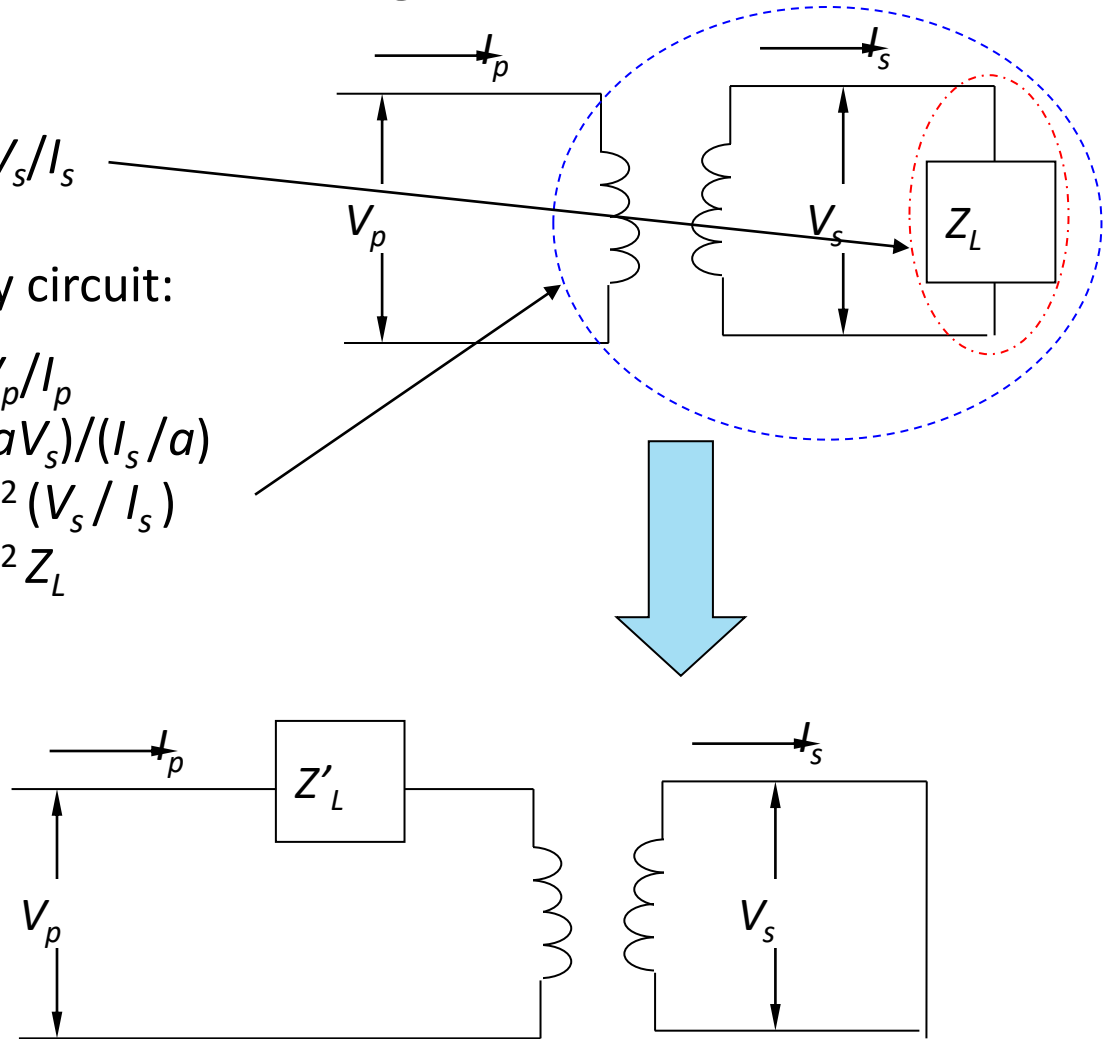
Impedance Transformation through a Transformer

Impedance of the load:

$$Z_L = V_s / I_s$$

The impedance of the primary circuit:

$$\begin{aligned} Z'_L &= V_p / I_p \\ &= (aV_s) / (I_s / a) \\ &= a^2 (V_s / I_s) \\ &= a^2 Z_L \end{aligned}$$



Equivalent Circuit with Converted Secondary-side Variables

Although the previous equivalent circuit is an accurate model of a transformer, it is not a very useful one. To analyze practical circuits containing transformers, it is normally necessary to convert the entire circuit to an equivalent circuit at a single voltage level. Therefore, the equivalent circuit must be referred either to its primary side or to its secondary side when solving the problems.

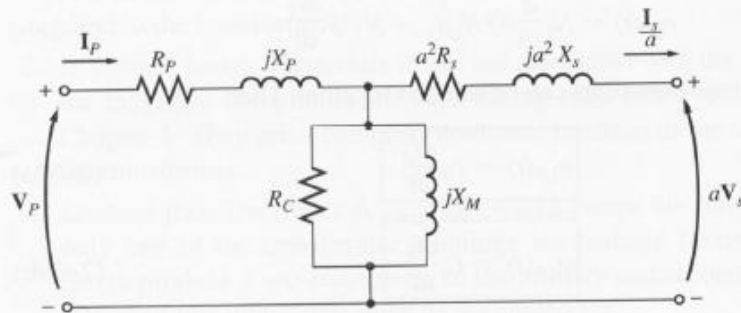


Figure (a) is the equivalent circuit of the transformer referred to its primary side.

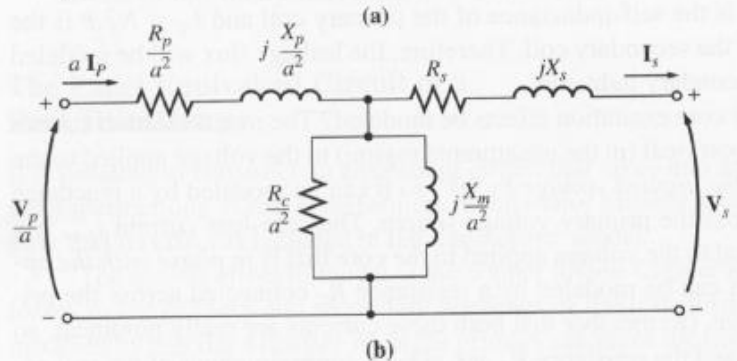


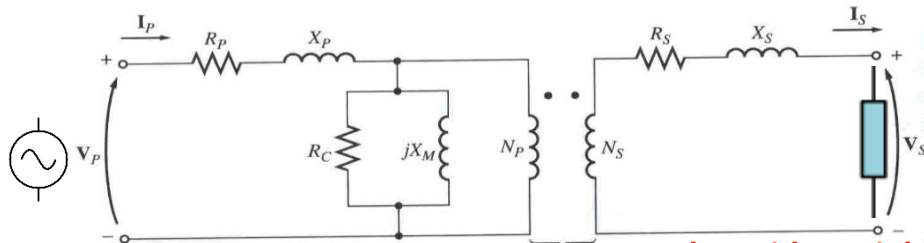
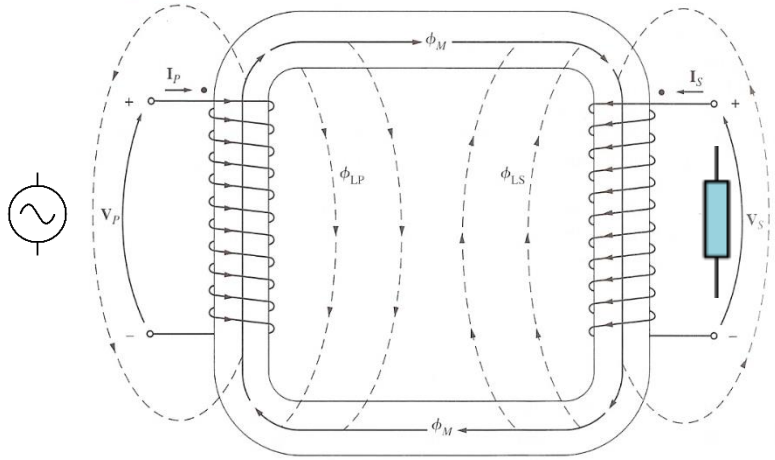
Figure (b) is the equivalent circuit referred to its secondary side.

The Equivalent Circuit

- To analyze practical circuits containing transformers, it is normally necessary to convert the entire circuit to an equivalent circuit at a single voltage level. Therefore, the equivalent circuit must be referred either to its primary side or to its secondary side in problem solutions.
 - a of Figure in the next slide is the equivalent circuit of the transformer referred to its primary side
 - b of Figure in the next slide is the equivalent circuit referred to its secondary side.

Summary of Circuit Model of Transformer

Figure 3-10 | Mutual and leakage fluxes in a transformer core.

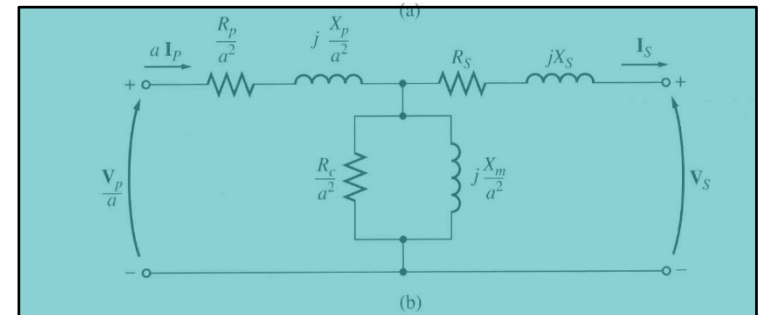
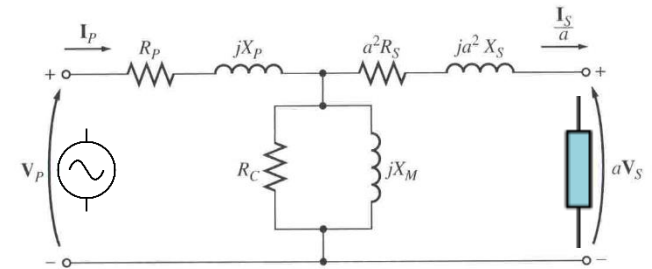


Primary side variables
(Actual)

Secondary side variables
(Actual)

Impedance of the load: $Z_L = V_s / I_s$

Equivalent impedance in the primary side : $Z'_L = V_p / I_p = (aV_s) / (I_s / a) = a^2 (V_s / I_s) = a^2 Z_L$

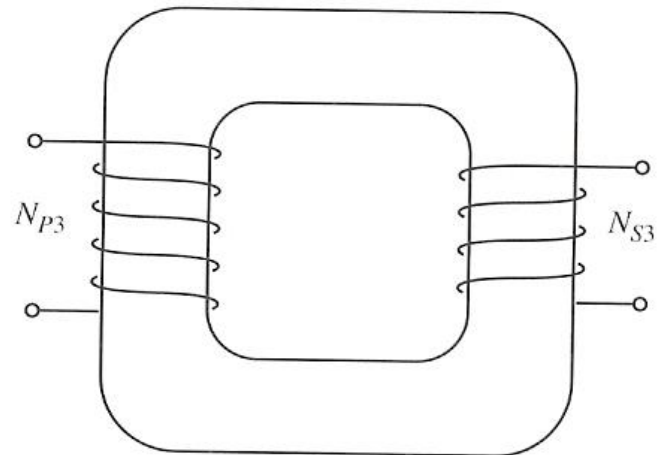
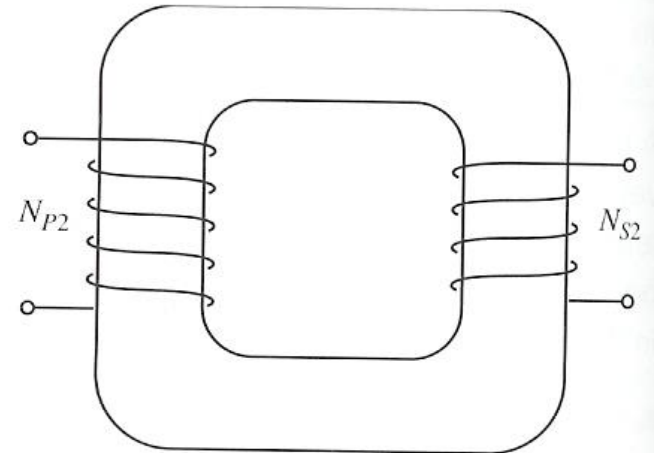
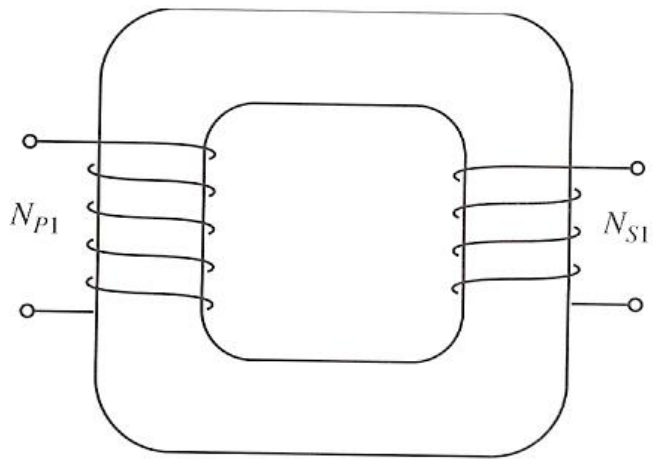


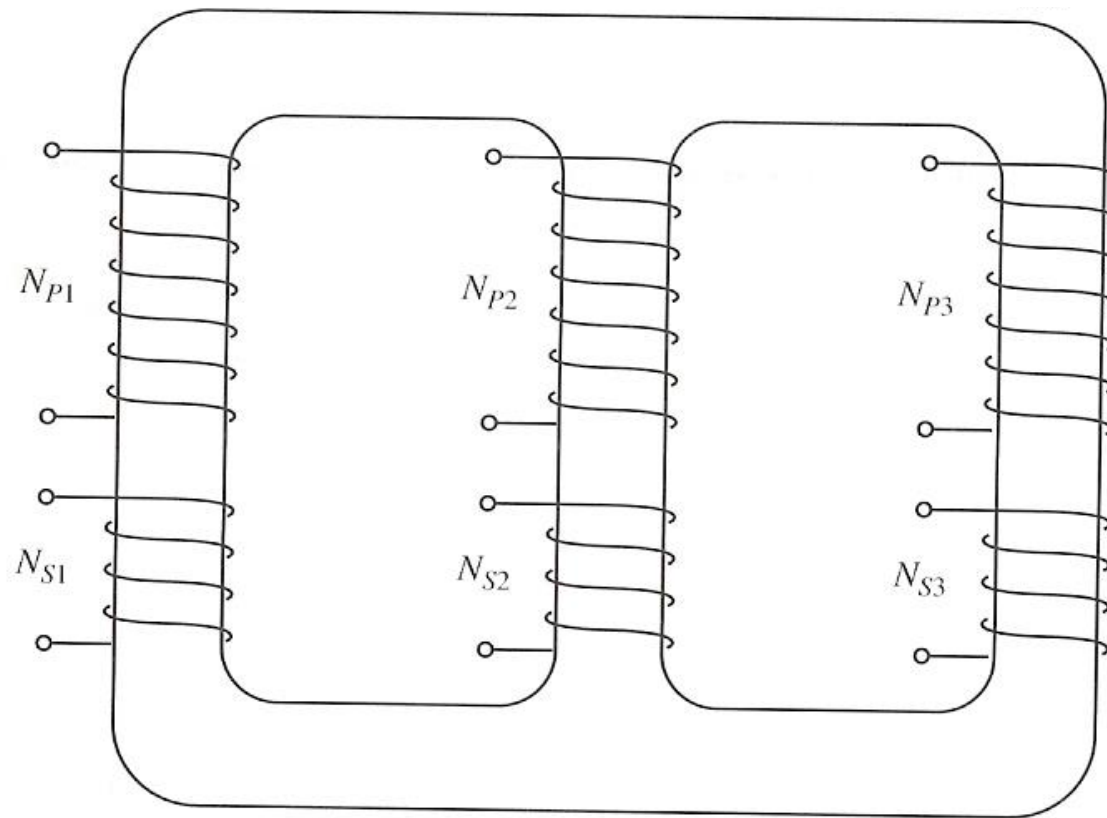
Extra materials:

Three Phase Transformer

Three-phase Transformers

- Two ways to construct a 3Φ Transformer
 - Take three 1Φ Transformers and connect them in a 3Φ bank.
 - Old type, but still in service
 - Each unit in the bank could be replaced individually.
 - Make a 3Φ transformer consisting of three sets of windings wrapped on a common core
 - Preferred in practice today
 - Lighter, smaller, cheaper, and slightly more efficient

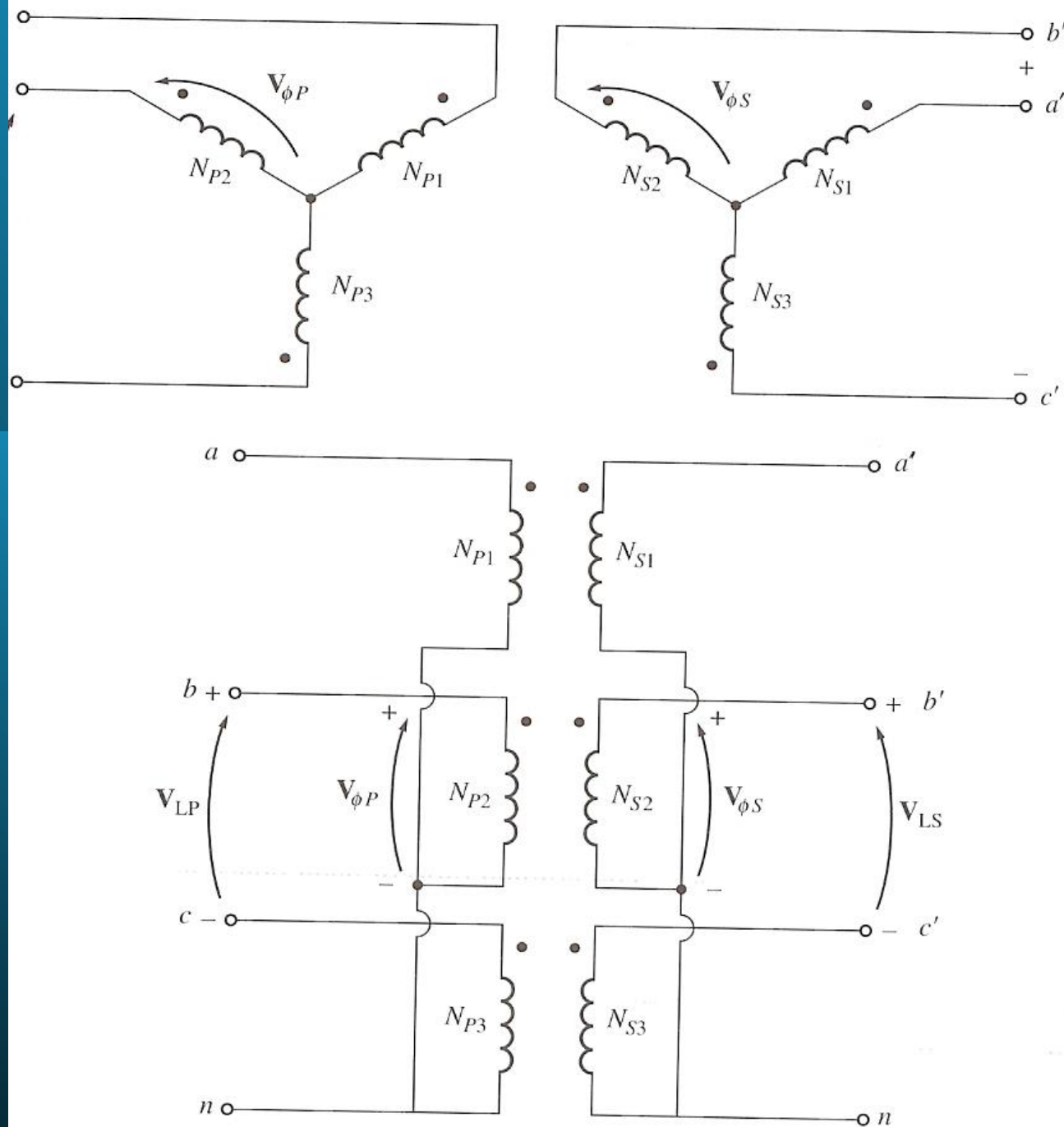




Three-phase Transformers Connection

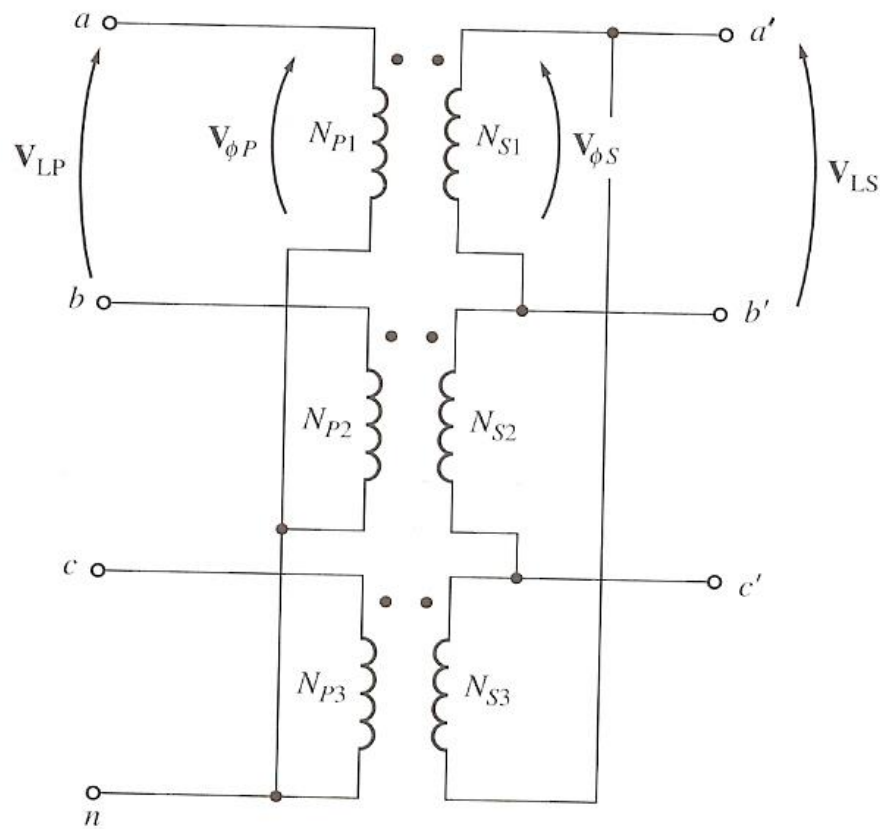
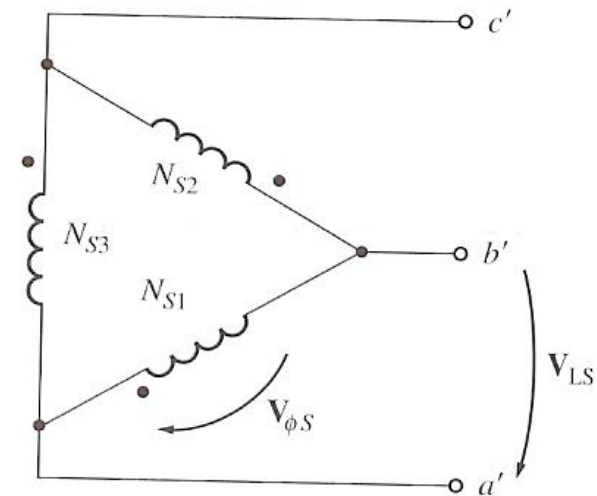
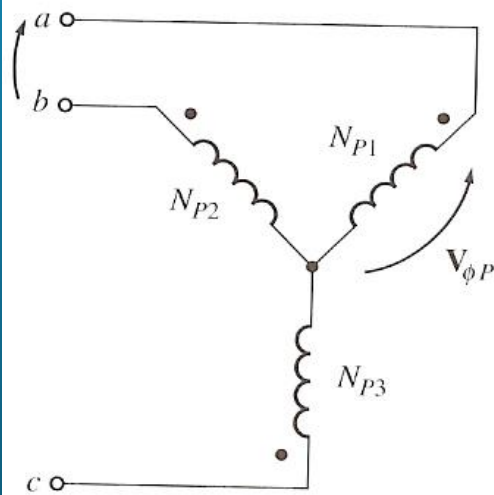
- 3 Φ Transformer Connections
 - Wye-wye (Y-Y)
 - Wye-delta (Y- Δ)
 - Delta-wye (Δ -Y)
 - Delta-Delta (Δ - Δ)
- The key to analyzing any 3 Φ transformer bank
 - Any 1 Φ transformer in the bank behaves exactly like the 1 Φ transformers already studied.
 - The impedance, voltage regulation, efficiency, and similar calculations for 3 Φ transformers are done on a per-phase basis, using exactly the same techniques already developed for 1 Φ transformers.

Wye-Wye (Y-Y) Connection



(a)

Wye-Delta (Y- Δ) Connection

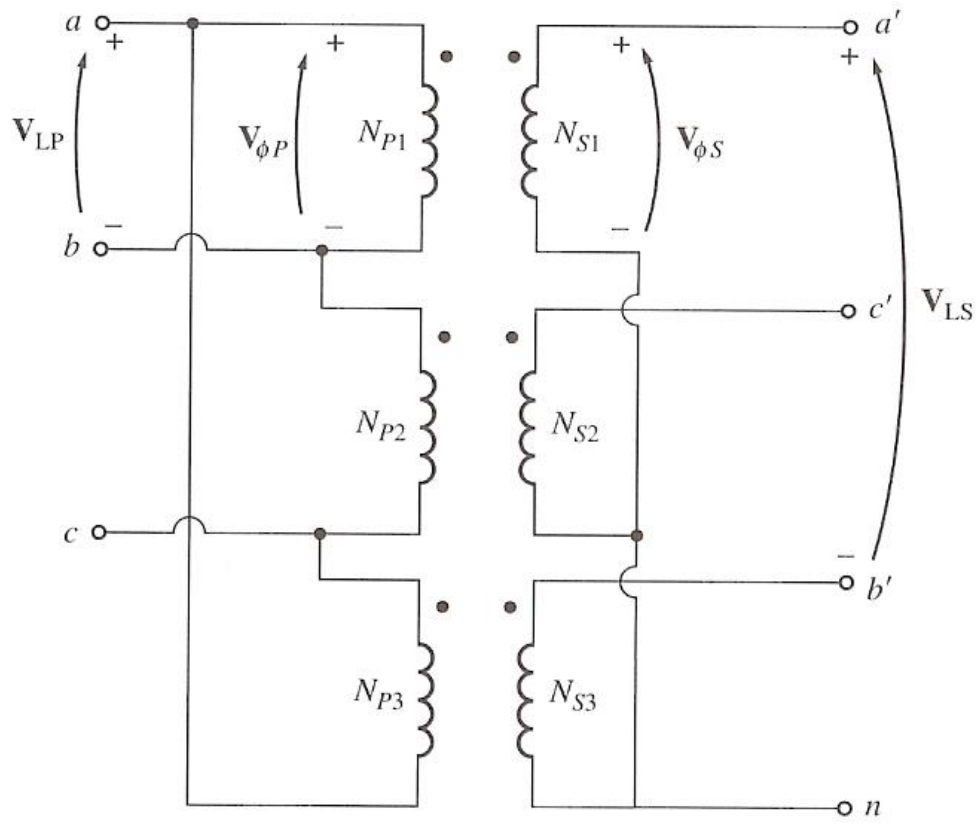
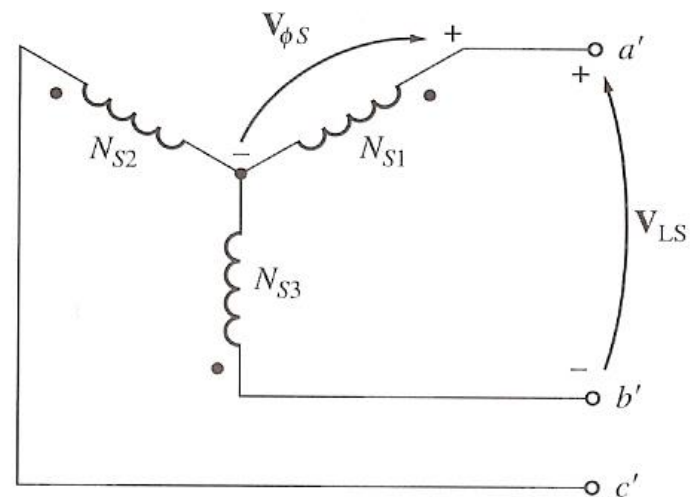
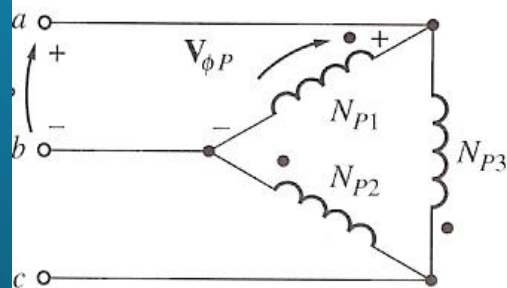


Three-phase Transformer –Cont.

- Wye-Delta (Y-Δ) Connection
 - Turns ratio :

$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{V_{\phi S}} = \sqrt{3}a \left(\frac{V_{\phi P}}{V_{\phi S}} = a \right)$$

Delta-Wye (Δ -Y) Connection

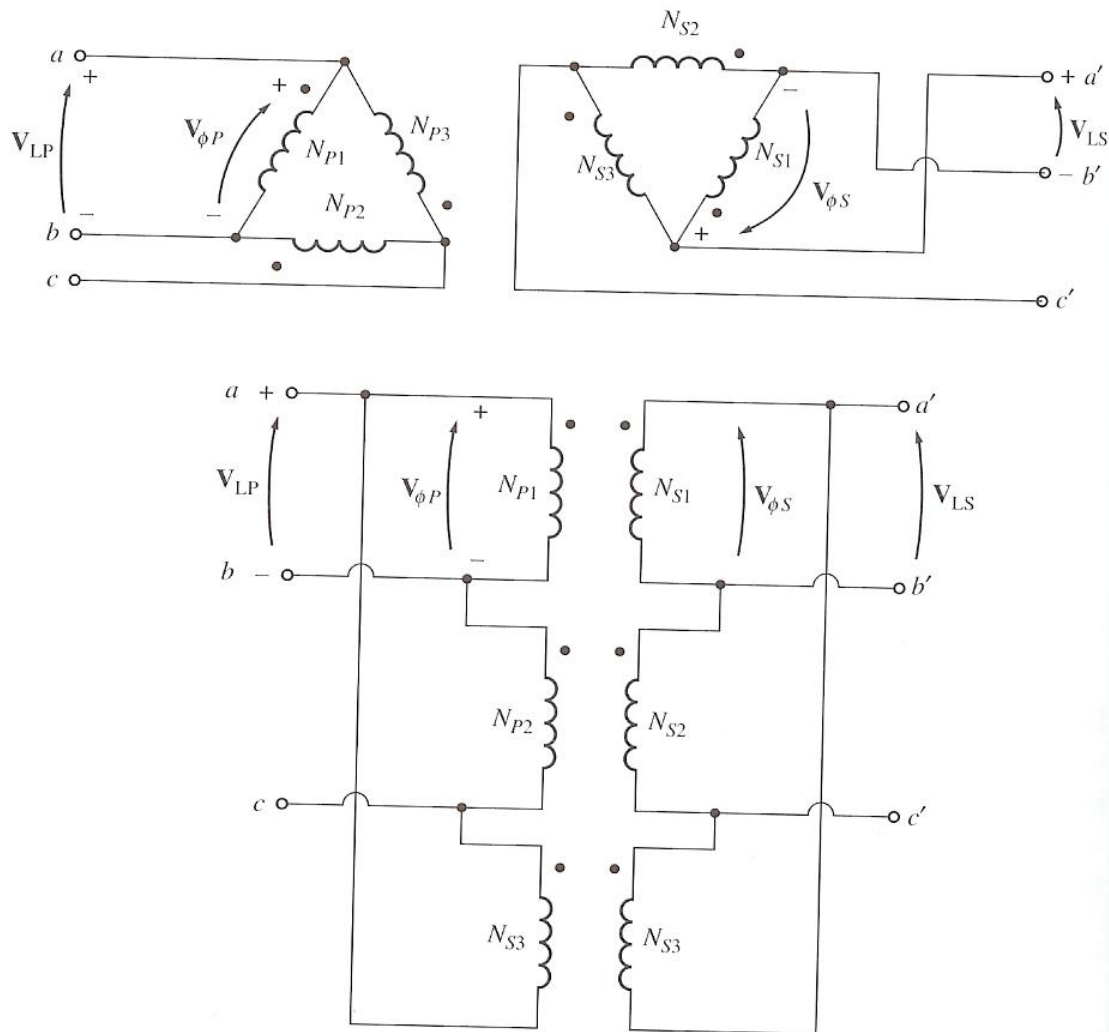


Three-phase Transformers –Cont.

- Delta-Wye (Δ -Y) Connection
 - Turns ratio :

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{\sqrt{3}V_{\phi S}} = \frac{a}{\sqrt{3}} \left(\frac{V_{\phi P}}{V_{\phi S}} = a \right)$$

Delta-Delta (Δ - Δ) Connection



Three-phase Transformers

– *Cont.*

- Delta-Delta (Δ - Δ) Connection
 - Turns ratio :

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{V_{\phi S}} = a$$